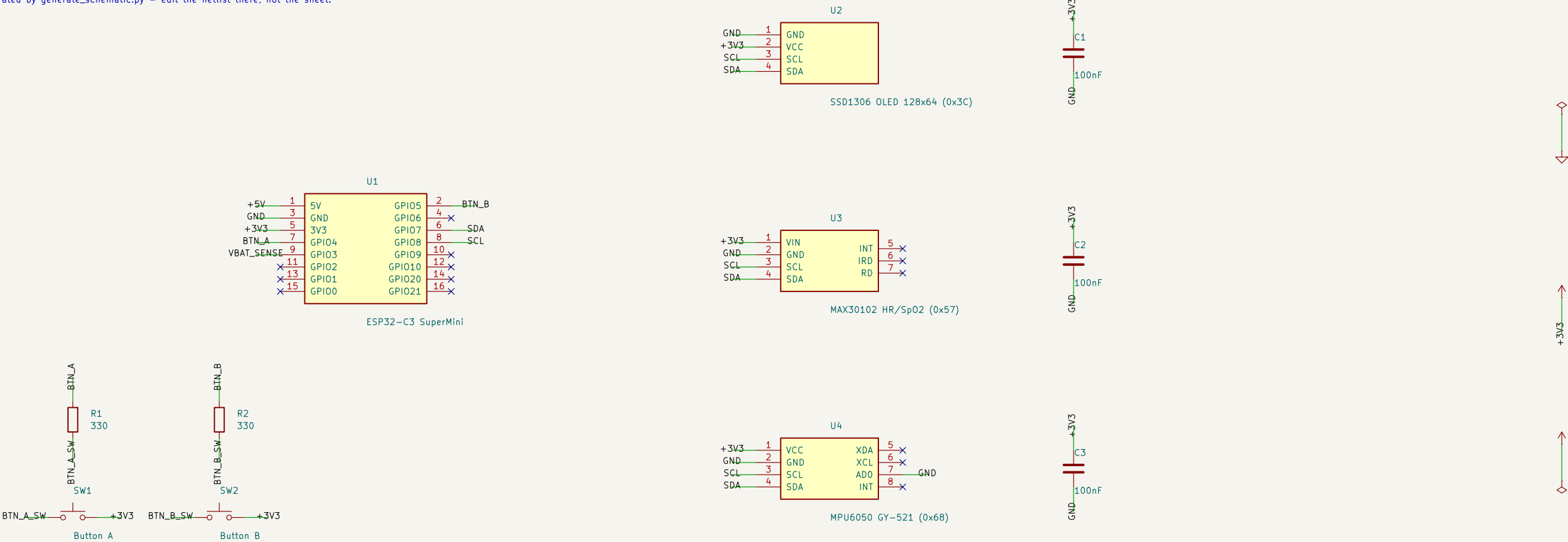


FitnessAI Watch – ESP32–C3 SuperMini + SSD1306 + MAX30102 + MPU6050

Connections are made by NET LABEL: identically–named labels are one net.

Pin assignment is the source of truth in watch/firmware/fitness\_watch/config.h.

Generated by generate\_schematic.py – edit the netlist there, not the sheet.



I2C runs at 100 kHz. Pull–ups are already on the breakout modules

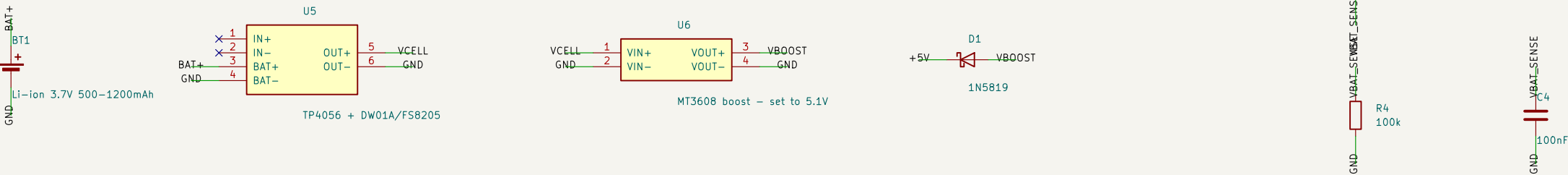
(~1.1k combined). Do not add more – see watch/README.md > Resistors.

POWER: the battery runs the watch unless USB–C is plugged in.

The SuperMini's 5V pad IS USB VBUS. USB present = 5.0V on that node; the

battery branch can only reach 5.1V – 0.35V (D1) = 4.75V, so D1 reverse–

biases and the battery drops out. Unplug USB and it conducts again.



TP4056 IN+/IN– = the charger module's OWN USB socket. Do NOT wire them to

the ESP32 5V pin: that makes a loop that charges the battery from itself

(boost → 5V → charger → cell → boost). It never terminates. See POWER.md.

SET THE MT3608 TO 5.1V, WITH NOTHING ON ITS OUTPUT, BEFORE WIRING IT UP.

These boards ship at an arbitrary setting and can put out up to 28V.

Sheet: /  
File: fitness\_watch.kicad\_sch

**Title:**

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.5

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